Revisiting Balassa–Samuelson: Theory, Evidence, and Measurement

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Long-Run Real Exchange Rates Are Hard to Explain

- Real exchange rates move a great deal over long horizons
- Leading explanation: Balassa–Samuelson
  - Faster traded productivity raises wages
  - Higher wages raise non-traded prices: the RER appreciates
- The evidence is mixed
  - Support in some old studies;
  - Weak or reversed in others (more recent)

▸ Literature

This Paper

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    - IO weights, export and import prices, coefficients of one

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2. Reduced-form evidence
   - Classic test, as commonly run, fails: partial $R^2 \approx 0$
   - The US is in every observation, so must add year effects: partial R^2 0.13–0.16
   - Modern test: partial R^2 0.18 for productivity, 0.57 for export prices

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   - The US is in every observation, so must add year effects: partial R^2 0.13–0.16
   - Modern test: partial R^2 0.18 for productivity, 0.57 for export prices
3. Quantification
   - Main results: trade wedges $R^2 = 0.61$; productivity and labor wedges ≈ 0
   - Unpacking trade wedges: “appeal” carries the RER result

Today's Plan

A Theory of Real Exchange Rates

Reduced-Form Evidence
- The Classic Test
- The Modern Test

Quantification
- Main Results
- Unpacking Trade Wedges

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A General Neoclassical Economy

Countries i; firms $n \in \mathcal{N}_i$; one local factor, labor; competitive markets

Firms:	$q_n \leq A_n f_n(\ell_n, m_n)$	CRS; inputs from any firm
Households:	$\max\ \tilde{c}_i(c_i)$ s.t. $p \cdot c_i \leq D_i$	any homothetic preferences
Labor:	$u_n w_n = \lambda_i$	wedge-adjusted wages equal λ_i

- Left open: spending D_i and labor supply ℓ_i, so any intertemporal model
- Trade costs: a shipping firm turns a good at home into the same good abroad
- $\mathrm{RER}_i = P_i / P_b$, with consumer prices chained as statistical agencies do

Two Cost Equations, One Wage Scale

$$\underbrace{d \log p_i^E}_{\text{export price}} = \overline{x}_i^{\ell E}\, d \log \lambda_i - \textstyle\sum_r \left(\overline{x}_{ri}^{E}\, d \log A_r + \overline{x}_{ri}^{\ell E}\, d \log u_r \right) + \sum_{r \notin i} \overline{x}_{ri}^{ME}\, d \log p_r$$

$$\underbrace{d \log P_i}_{\text{CPI}} = \overline{x}_i^{\ell C}\, d \log \lambda_i - \textstyle\sum_r \left(\overline{x}_{ri}^{C}\, d \log A_r + \overline{x}_{ri}^{\ell C}\, d \log u_r \right) + \sum_{r \notin i} \overline{x}_{ri}^{MC}\, d \log p_r$$

- Superscripts: E exports, C consumption; ℓ local labor, M imports
- $\overline{x}_i^{\ell E}$: share paid to local labor; $\overline{x}_{ri}^{\ell E}$: to labor at firm r
- $\overline{x}_{ri}^{E}$: Domar weight of firm r, directly and through suppliers
- $\overline{x}_{ri}^{ME}$: share spent on imports from foreign firm r
- A_r, u_r: productivity and labor wedges, the model's primitives
- p_r, $r \notin i$: prices of imported goods, set in world equilibrium

The CPI Follows Export Prices and Export Productivity

$$d \log \mathrm{RER}_i = \overline{x}_i^{\ell C} \left(d \log \overline{P}_i^E + d \log \overline{A}_i + d \log \overline{u}_i - d \log \overline{P}_i^M \right) - \overline{x}_b^{\ell C} \left(d \log \overline{P}_b^E + d \log \overline{A}_b + d \log \overline{u}_b - d \log \overline{P}_b^M \right)$$

$$d \log \overline{P}_i^E = \frac{d \log p_i^E}{\overline{x}_i^{\ell E}}, \qquad d \log \overline{A}_i = \sum_r \left(\frac{\overline{x}_{ri}^E}{\overline{x}_i^{\ell E}} - \frac{\overline{x}_{ri}^C}{\overline{x}_i^{\ell C}} \right) d \log A_r$$

- Eliminate λ_i between the two equations; b: the benchmark country
- $\overline{x}_i^{\ell C}$, $\overline{x}_i^{\ell E}$: shares of consumption and of exports paid to local labor
- $\overline{x}_{ri}^C$, $\overline{x}_{ri}^E$: Domar weights of firm r in consumption and in exports
- p_i^E: export price; $\overline{P}_i^E$: export price rescaled by $1/\overline{x}_i^{\ell E}$; P_i: CPI
- $\overline{u}_i$, $\overline{P}_i^M$: same form, with labor shares and import shares

Classic Balassa–Samuelson Is a Special Case

$$\log \mathrm{RER}_{it} = \kappa_{ib} + s_N \left(z_{it} - z_{bt} \right), \qquad z_{it} \equiv \log \left(\frac{A_{Tit}}{A_{Nit}} \right)$$

- No input–output linkages; labor is the only factor
- One traded and one non-traded good per country
- Traded goods are perfect substitutes with a common price
- s_N: expenditure share on the non-traded good
- Traded productivity then moves wages one for one

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Data

- RER: PWT consumption price levels; benchmark United States
- Productivity: real value added per employed person
  - EU KLEMS 2023 (28), CIP 4.0 (China), OECD STAN (6)
- 35 economies; all regressions end in 2017
- Classic: traded = agriculture, mining and manufacturing
- Modern: 16 sectors, 8 within manufacturing
  - OECD input–output tables and supplier prices

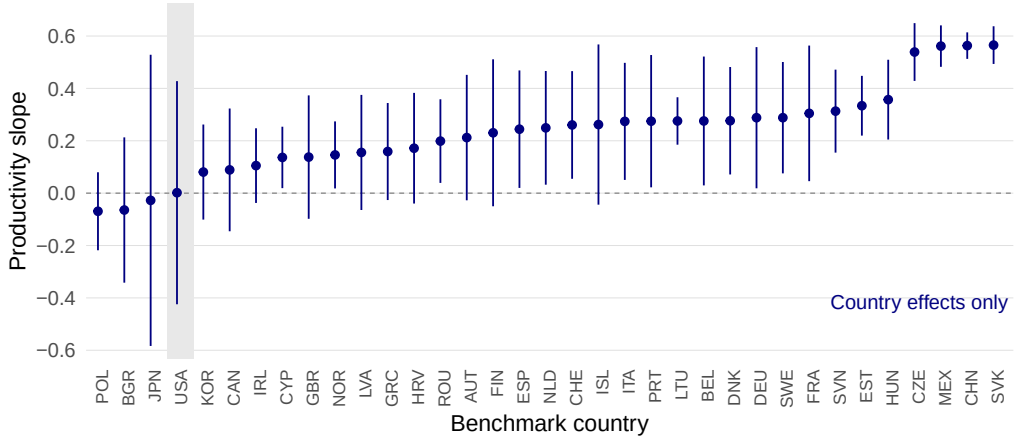
The Classic Test as Commonly Run Fails

$$\log \mathrm{RER}_{it} = \delta \left(z_{it} - z_{bt} \right) + \psi_i + \varepsilon_{it}$$

	1970–2017	2000–2017
Productivity	0.10	0.00
	(0.08)	(0.21)
Partial R^2	0.01	0.00
Observations	1,182	612

- Country effects only; benchmark b: the United States
- Theory predicts $\delta = s_N > 0$; estimates are small and insignificant

Even worse, the Answer Depends on the Benchmark



- Same regression, 2000–2017, each country in turn as benchmark (US shaded)
- Country effects only: -0.07 to 0.57, significant for 21 of 35

Issue: The Benchmark Is in Every Observation

Common practice: $\log \mathrm{RER}_{it} = \delta\,(z_{it} - z_{bt}) + \psi_i + \varepsilon_{it}$

This paper: $\log \mathrm{RER}_{it} = \delta z_{it} + \psi_i + \psi_t + \varepsilon_{it}$

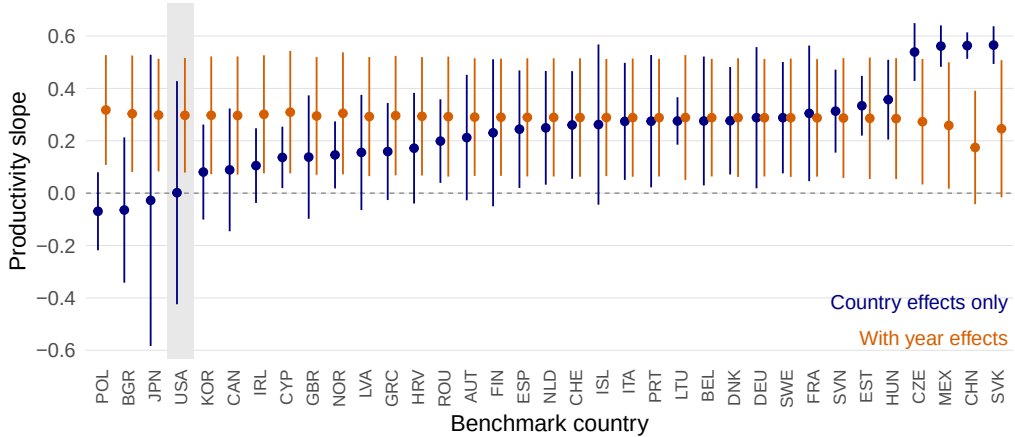
- US prices and productivity enter every observation
- A US shock or measurement error hits every country at once
- Adding countries (adding N) does not average it out
- Year effects absorb it; theory still predicts $\delta = s_N$

With Year Effects, the Classic Test Works

	Country effects		Country and year effects	
	1970–2017	2000–2017	1970–2017	2000–2017
Productivity	0.10	0.00	0.22	0.30
	(0.08)	(0.21)	(0.06)	(0.11)
Partial R^2	0.01	0.00	0.13	0.16
Observations	1,182	612	1,182	612

- Right sign: 0.22 and 0.30, significant at 5%
- Balassa-Samuelson is rescued!
- But small: well below the non-traded share

With Year Effects, the Benchmark Barely Matters



- Country effects only: -0.07 to 0.57, significant for 21 of 35
- With year effects: 0.17 to 0.32, significant for 33 of 35

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The Modern Test: IO Weights and Trade Prices

$$\log \mathrm{RER}_{it} = \psi_i + \psi_t + \delta \left(X_{it}^A + X_{it}^E + X_{it}^M \right) + \varepsilon_{it}$$

- X_{it}^A: sectoral productivity, weighted by IO exposure
- X_{it}^E: export prices of domestic suppliers
- X_{it}^M: signed import prices of foreign suppliers
- Cumulated from 2000; the 612 country-years of the classic test
- Predicted coefficient: one

▸ IO index

IO Weights Help a Little; Export Prices Fit Best

	Productivity	Export price	Import price	All three
Annual levels, country and year effects				
Coefficient	0.28	0.88	0.04	0.41
	(0.06)	(0.12)	(0.66)	(0.05)
Partial R^2	0.18	0.57	0.00	0.54

IO Weights Help a Little; Export Prices Fit Best

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Annual levels, country and year effects				
Coefficient	0.28	0.88	0.04	0.41
	(0.06)	(0.12)	(0.66)	(0.05)
Partial R^2	0.18	0.57	0.00	0.54
Long difference, 2000–2017				
Coefficient	0.26	0.75	−0.76	0.33
	(0.07)	(0.16)	(1.31)	(0.08)
R^2	0.32	0.50	0.01	0.62

- Productivity: partial R^2 of 0.18, against 0.16 classic
- Export prices: 0.57, with a coefficient of 0.88

Robust across Prices and Samples

	Export price		Productivity	
	Coefficient	Partial R^2	Coefficient	Partial R^2
OECD dollar prices	0.88 (0.12)	0.57	0.28 (0.06)	0.18
PWT trade deflators	0.78 (0.15)	0.44	0.28 (0.06)	0.18
IO weights fixed at 2000	0.81 (0.13)	0.58	0.28 (0.11)	0.11
Euro area and euro pegs	0.55 (0.12)	0.40	0.17 (0.10)	0.05
Local currency	0.69 (0.07)	0.57	0.08 (0.09)	0.02

- Euro pegs: year effects absorb the common dollar rate
- Local currency: no exchange rate on either side

▸ PWT ▸ Fixed weights ▸ Euro ▸ Local currency

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Endogenous Prices: A Multi-Sector Armington Model

Across sectors:	$C_j = \left[\sum_s \beta_{js}^{1/\eta} \, C_{js}^{(\eta-1)/\eta} \right]^{\eta/(\eta-1)}$	$\eta = 0.6$: complements
Within a sector:	$C_{js} = \left[\sum_i C_{ijs}^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}$	$\sigma = 4$: across origins
Production:	$p_{is} = \left(\frac{w_{is}}{A_{is}} \right)^{\alpha_{is}} \prod_r P_{ir}^{\gamma_{irs}}$	labor and IO inputs
Delivery:	$p_{ijs} = T_{ijs} \, p_{is}$	trade wedge, $T_{iis} = 1$

- 35 economies and the rest of the world; 16 sectors; 2000–2017
- Wages, export and import prices are equilibrium outcomes

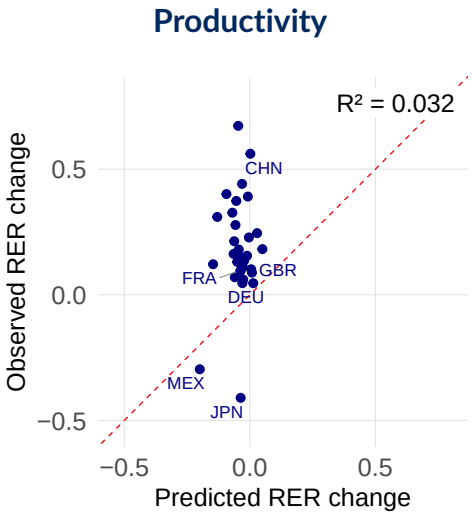
Three Primitives, Shocked from 2000 to 2017

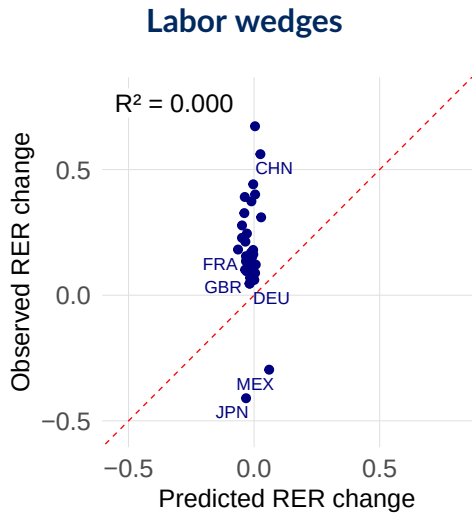
- Productivity A_{is}: measured real value added per worker
- Labor wedges u_{is}, with $u_{is} w_{is} = \lambda_i$
  - Fed in: relative sectoral wages, $\Delta \log u_{is} = -\Delta \log w_{is}$ up to a country constant
  - Not fed in: the national wage λ_i, which the model solves
- Trade wedges T_{ijs}: trade costs and appeal, matching observed trade shares

$$T_{ijs} = \frac{p_{js}}{p_{is}} \left(\frac{\pi_{ijs}}{\pi_{jjs}} \right)^{1/(1-\sigma)}$$

- Shock each alone, then all three; compare with observed RER changes
  - 33 countries with sectoral wage data, relative to the US

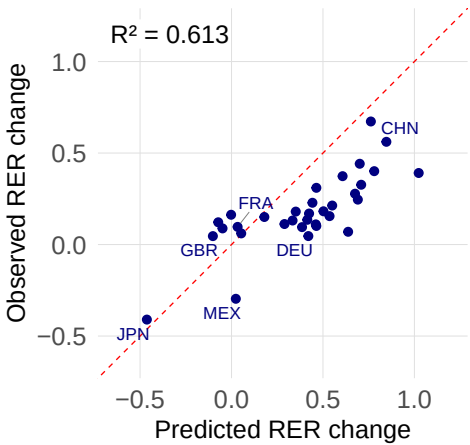
Productivity and Labor Wedges Predict Nothing



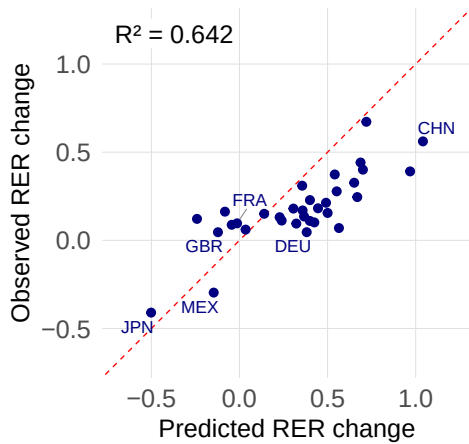


Trade Wedges Predict RER Changes

Trade wedges



All three



Why Productivity and Labor Wedges Barely Matter

- Productivity lowers export prices
  - Wages rise less than productivity; at $\sigma = 4$, prices undo most gains
- Labor wedges only move relative wages within a country
  - A common shift is absorbed by the wage scale λ_i
- Trade wedges move demand for a country's exports; wages follow

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Trade Wedges: Trade Costs or Appeal?

$$T_{ijs} = \underbrace{\tau_{ijs}}_{\text{trade cost}} \times \underbrace{\left(\frac{\phi_{is}}{\phi_{js}}\right)^{\frac{1}{1-\sigma}}}_{\text{relative appeal}}$$

- Trade cost τ_{ijs}: cost of shipping from i to j; $\tau_{iis} = 1$
- Appeal ϕ_{is}: demand for i's goods at given prices, at home and abroad
- Which one moved real exchange rates? Three exercises:
  1. Symmetric trade costs, inferred from trade flows
  2. Freight and tariffs, measured directly
  3. Appeal, fitted after removing freight and tariffs

Exercise 1: Measuring Symmetric Trade Costs

$$h_{ijs} = \frac{1}{2(1-\sigma)} \Delta \log \left(\frac{X_{ijs} X_{jis}}{X_{iis} X_{jjs}} \right) \qquad X_{ijs}\text{: sales from } i \text{ to } j$$

How we measure it

- Head–Ries index of the trade cost between i and j
- Appeal and prices raise i's sales everywhere alike, so they cancel
- Trade flows alone measure it, in 2000 and in 2017

How it changed, 2000–2017

- Small: SD 0.48 across links, against 2.61 for all wedge changes

▸ Asymmetric part ▸ Country effects

Exercise 1: Symmetric Costs Do Not Move RERs

The experiment

- Change wedges by h_{ijs} only
- Everything else at 2000

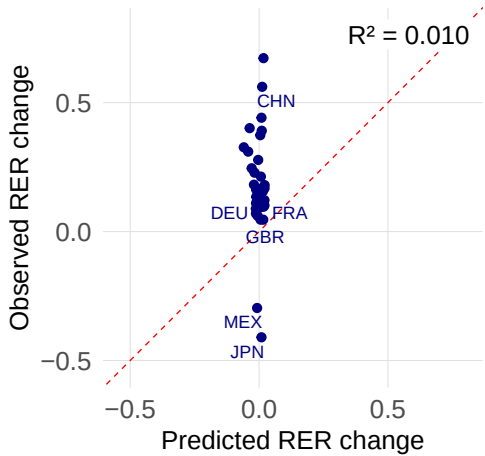
Exercise 1: Symmetric Costs Do Not Move RERs

The experiment

- Change wedges by h_{ijs} only
- Everything else at 2000

What it implies

- $R^2 = 0.010$
- Cheaper exports appreciate
- Cheaper imports depreciate
- The two offset



Exercise 2: Measuring Freight and Tariffs

$$(1-\sigma)\Delta \log T_{ijs} = \Delta \log \phi_{is} - \Delta \log \phi_{js} \underbrace{- \sigma \, \Delta \log(\text{freight} \times \text{duty})_{ijs}}_{\text{measured costs}} + \text{unmeasured}$$

How we measure it

- Freight: OECD cif–fob margins; tariffs: WITS applied rates; 2000 and 2017
- Exponent $-\sigma$: trade flows are valued before freight and duties

How they changed, 2000–2017

- Small: at the median link, freight -0.9%, duties unchanged
- SD across links 0.16, against 2.61 for all wedge changes
- Import costs fell more than export costs, -3.1% against -1.6%
  - Tariff cutters: Mexico -13% against -1%; Bulgaria -11% against -2%

Exercise 2: Freight and Tariffs Do Not Move RERs

The experiment

- Change wedges by freight, duties
- Everything else at 2000

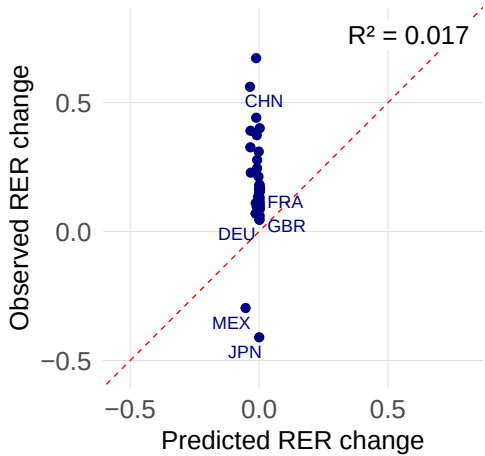
Exercise 2: Freight and Tariffs Do Not Move RERs

The experiment

- Change wedges by freight, duties
- Everything else at 2000

What it implies

- $R^2 = 0.017$
- Freight 0.027; duties 0.011
- Tariff cutters: imports cheaper
- So they are pushed to depreciate



Exercise 3: Measuring Appeal

$$\underbrace{(1-\sigma)\Delta \log T_{ijs} + \sigma\, \Delta \log(\text{freight} \times \text{duty})_{ijs}}_{\text{wedge net of measured costs}} = \Delta \log \phi_{is} - \Delta \log \phi_{js} + \text{unmeasured}$$

How we measure it

- Net measured freight and duties out of each wedge change
- Fit $\Delta \log \phi_{is}$ by sector, with 2000 trade weights
- What appeal cannot fit is the unmeasured trade cost
- One-sided non-tariff barriers stay in appeal

How it changed, 2000–2017

- Large: SD 2.56 across links, against 2.61 for all wedge changes
- Unmeasured costs are small: SD 0.51

Exercise 3: Appeal Explains RER Changes

The experiment

- Change wedges by appeal only
- Everything else at 2000

▸ Table

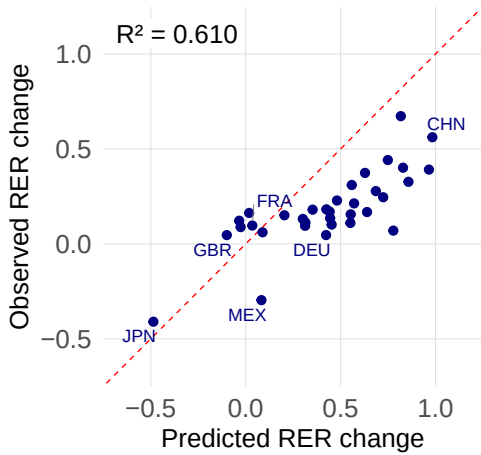
Exercise 3: Appeal Explains RER Changes

The experiment

- Change wedges by appeal only
- Everything else at 2000

What it implies

- $R^2 = 0.610$; all wedges 0.621
- Unmeasured costs: $R^2 = 0.002$
- Appeal raises demand for i's goods
- So its wages and CPI rise



▸ Table

Check 1: Model Export Prices Move Like Actual Ones

The question

- Could appeal be a price error?

What we do

- Model versus OECD export prices
- OECD prices are never targeted

▸ Prices and RERs

Check 1: Model Export Prices Move Like Actual Ones

The question

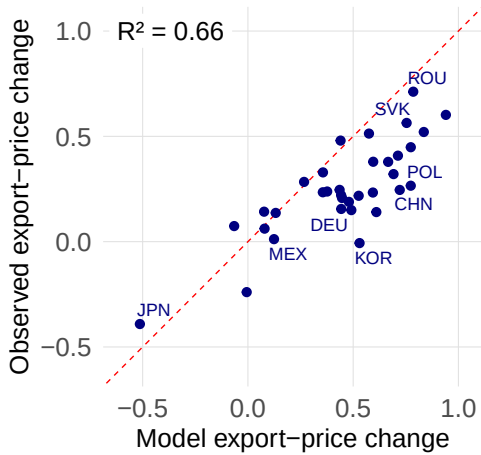
- Could appeal be a price error?

What we do

- Model versus OECD export prices
- OECD prices are never targeted

What we find

- Across countries: $R^2 = 0.66$
- Model changes are larger: slope 0.59



▸ Prices and RERs

Check 2: Appeal Rebuilt from Observed Prices Agrees

The question

- Does appeal look the same when built from observed prices?

What we do

1. Trade flows show which exporter–sectors gained sales: $\hat{e}_{is}$
2. Remove gains due to observed price changes: $\hat{e}_{is} + (\sigma - 1)\Delta \log p_{is}$
3. Compare with the model's appeal, within each country

▸ Details ▸ Slopes

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3. Compare with the model's appeal, within each country

What we find

- Partial $R^2 = 0.40$, with or without export weights
- Sales gains alone: 0.12 to 0.19
- It tests appeal's pattern across sectors within a country

▸ Details ▸ Slopes

Check 3: Comparing Appeal with Export Quality

The question

- Is appeal the quality of what a country exports?

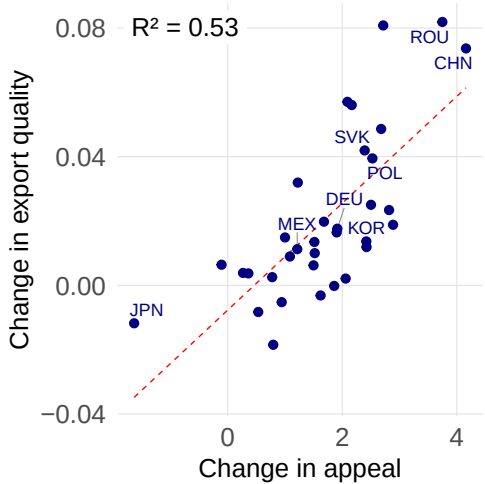
What we do

- Appeal: change in a country's manufacturing appeal, 2000–2014
- Export quality: IMF estimates from export unit values, 2000–2014
- Unit value $=$ quality $\times$ quality-adjusted price
- Correlate these changes across 34 countries, relative to the US

Check 3: Appeal Moves with Export Quality

What we find

- Higher appeal, higher export quality
- Quality explains half: $R^2 = 0.53$



▸ Prices and RER ▸ Table

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- The classic test needs year fixed effects
  - Otherwise, the results are fragile
  - With them, right sign but small size
- Wages and the CPI follow export prices, as Balassa–Samuelson says
  - Export prices account for most RER changes, in data and in the model
- Export prices moved with “appeal”
  - Appeal moves with measured export quality
- Next: what is “appeal”, really?
  - Product-level trade data?
  - New products, quality, and non-tariff barriers

Appendix

Related Literature

- Panel tests: De Gregorio et al. (1994), Canzoneri et al. (1999)
- Mixed evidence: Lee and Tang (2007), Gubler and Sax (2019)
- Labor wedges: Berka et al. (2018), Devereux et al. (2025)
- Traded goods prices: Engel (1999), Betts and Kehoe (2006)
- Terms of trade: Benigno and Thoenissen (2003)
- Accounting with wedges: Chari et al. (2007), Eaton et al. (2016)

Back

IO Productivity Index

$$\Delta_t z_{it}^{\mathrm{IO}} = \frac{1}{2} \sum_{s=t-1}^{t} \sum_{r \in \mathcal{N}_i} \left(\frac{\overline{x}_{is}^{\ell C}}{\overline{x}_{is}^{\ell E}} \overline{x}_{ri,s}^{E} - \overline{x}_{ri,s}^{C} \right) \Delta_t \log A_{rt}$$

- Export exposure less consumption exposure, from IO tables
- Adjacent-year averages of the complete annual weights
- Cumulated from zero in 2000

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Modern Test: PWT Trade Deflators

	Productivity	Export price	Import price	All three
Annual levels, country and year effects				
Coefficient	0.28	0.78	0.06	0.37
	(0.06)	(0.15)	(0.81)	(0.05)
Partial R^2	0.18	0.44	0.00	0.45
Long difference, 2000–2017				
Coefficient	0.26	0.65	−0.23	0.30
	(0.07)	(0.18)	(1.26)	(0.08)
R^2	0.32	0.43	0.00	0.56

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Modern Test: IO Weights Fixed at 2000

	Productivity	Export price	Import price	All three
Annual levels, country and year effects				
Coefficient	0.28	0.81	−0.33	0.48
	(0.11)	(0.13)	(1.00)	(0.05)
Partial R^2	0.11	0.58	0.00	0.54
Long difference, 2000–2017				
Coefficient	0.26	0.68	−1.18	0.39
	(0.08)	(0.13)	(1.78)	(0.08)
R^2	0.21	0.55	0.02	0.62

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Modern Test: Euro Area and Euro Pegs

	Productivity	Export price	Import price	All three
Annual levels, country and year effects				
Coefficient	0.17	0.55	0.11	0.33
	(0.10)	(0.12)	(0.57)	(0.08)
Partial R^2	0.05	0.40	0.00	0.35

- 343 country-years from 22 countries
- Coarse code 1 with the euro as anchor in all twelve months

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Modern Test: Local Currency

	Productivity	Export price	Import price	All three
Annual levels, country and year effects				
Coefficient	0.08	0.69	1.97	0.37
	(0.09)	(0.07)	(0.97)	(0.09)
Partial R^2	0.02	0.57	0.09	0.41
Long difference, 2000–2017				
Coefficient	0.08	0.74	3.66	0.38
	(0.07)	(0.06)	(1.36)	(0.13)
R^2	0.02	0.70	0.16	0.46

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Trade Costs and Appeal: Details

	SD across goods links	R^2
All trade wedges	2.61	0.621
Symmetric trade costs (Head–Ries)	0.48	0.010
Measured trade costs: freight and duties	0.16	0.017
Freight	0.03	0.027
Duties	0.15	0.011
Unmeasured trade costs	0.51	0.002
Appeal	2.56	0.610

	Corr. with measured part	SD of measured part / SD
Export plus import effect	0.50	0.22
Export less import effect	-0.22	0.06

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The Asymmetric Part Is a Country–Sector Shift

$$\Delta \log T_{ijs} = \underbrace{h_{ijs}}_{\text{symmetric}} + \underbrace{\tfrac{1}{2}\left(d_{is} - d_{js}\right)}_{\text{country–sector shift}} + \underbrace{r_{ijs}}_{\text{residual}}$$

What we do

- h_{ijs}: the symmetric trade cost of Exercise 1
- Fit d_{is} to the rest of each wedge change, with 2000 trade weights
- d_{is}: how i's export wedges moved against its import wedges
- Move each part alone in the model

What we find

- d: 95.5% of the variance; $R^2 = 0.618$, as much as all trade wedges
- h: $R^2 = 0.010$; r: $R^2 = 0.004$

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Country Effects Have the Shape of Appeal

$$(1-\sigma)\Delta \log T_{ijs} = \text{export effect of } i + \text{import effect of } j + \text{error}$$

What we do

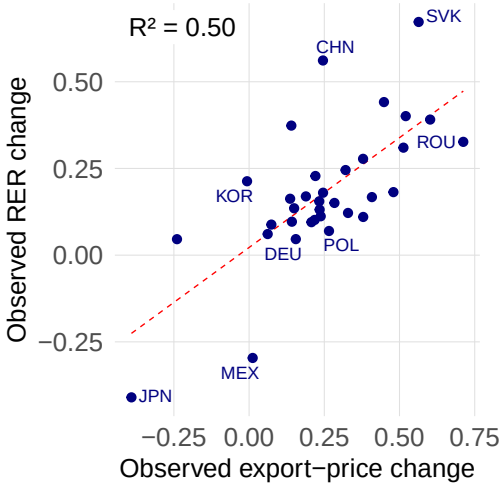
- One export and one import effect per country, by sector
- Appeal makes a country's two effects equal and opposite
- Costs falling on both of its sides move the two effects together

What we find

- Forcing the two to sum to zero: R^2 falls only from 0.96–0.99 to 0.92–0.97
- Measured costs: correlation 0.50 with the sum, -0.22 with the difference

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Observed Export Prices Track RERs



- $R^2 = 0.50$; slope 0.63
- 2000–2017 changes, 34 countries
- Prices the model never sees

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Appeal from Observed Prices: Details

	Weighted	Unweighted
Appeal from observed prices, $\hat{e}_{is} + (\sigma - 1)\Delta \log p_{is}$	0.40	0.40
Appeal from observed prices at the IV slope	0.11	0.18
Exporter-sector effects $\hat{e}_{is}$ alone	0.12	0.19
Exporter-sectors		560
Countries		35

- Partial R^2 with the model's appeal, net of country and sector means

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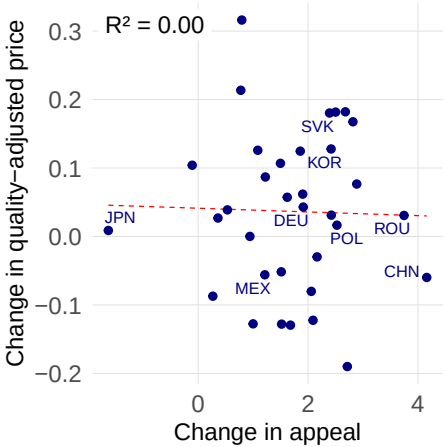
CDK Regressions

	Weighted		Unweighted	
	Coef.	S.e.	Coef.	S.e.
Bilateral flows on price change, OLS	0.10	0.18	-0.39	0.19
Exporter-sector effects on price change, OLS	0.11	0.26	-0.31	0.23
First stage: price change on productivity growth	-0.40	0.06	-0.30	0.03
Reduced form: effects on productivity growth	-0.02	0.13	0.30	0.09
Effects on price change, IV	0.06	0.31	-1.00	0.30
First-stage F	50		91	
Bilateral links / exporter-sectors	19,451 / 560			

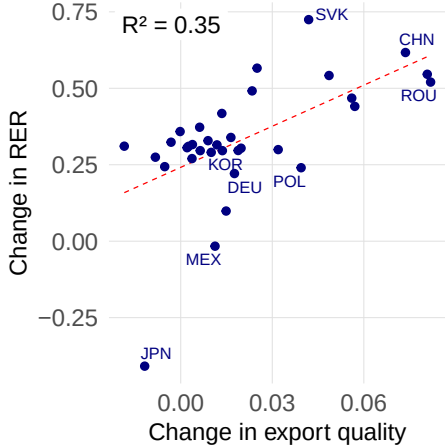
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Check 3: Quality-Adjusted Prices and the RER

Appeal and quality-adjusted prices



Export quality and the RER



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Appeal and Export Quality, through 2017

	Manufacturing appeal		
	2000–2014	2000–2017	RER, 2000–2014
Manufacturing quality	0.53	0.48	0.35
All-products quality	0.44	0.29	
Manufacturing unit value	0.02	0.01	0.10
Quality-adjusted unit value	0.00	0.01	0.03
RER	0.52	0.41	
Countries		34	

- Each cell: R^2 between the row and column changes
- 2000–2017 appeal: fit net of measured costs, as in Exercise 3

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